

SCOTT NEVILLE

CURRICULUM VITAE

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ACADEMIC INTERESTS

Main interests: Quiver mutations, algebraic and algorithmic combinatorics.

Current projects: Mutation equivalence and mutation invariants of quivers.

Previous projects: Theoretical and applied machine learning, polymer growth, algorithms in number theory.

EDUCATION

University of Utah, B.S. in Computer Science and Mathematics, 2017.

Senior thesis: The complexity of explaining neural networks through (group) invariants.

University of Cambridge, Masters of Advanced Studies in Mathematics (Part III), 2019.

Final Project: Mestre's algorithm for hyperelliptic curves.

University of Michigan, Ph.D. in Mathematics, 2025.

Thesis: [Combinatorics of quiver mutations](#). Advisor: Sergey Fomin.

PUBLICATIONS AND PREPRINTS

1. [An \$m\$ -ary partition generalization of a past Putnam problem](#) (with T. B. Flowers and J. A. Sellers), *Australas. J. Combin.* **72** (2018), 369–375.
2. [The complexity of explaining neural networks through \(group\) invariants](#) (with D. Ensign, A. Paul, and S. Venkatasubramanian), *Proc. Mach. Learn. Res., Algorithmic Learning Theory* **76** (2017), 341–359. Extended version, *Theoret. Comput. Sci.* **808** (2020), 74–85.
3. [Runaway feedback loops in predictive policing](#) (with D. Ensign, S. A. Friedler, C. Scheidegger, and S. Venkatasubramanian), *Proc. Conf. Fairness, Accountability, and Transparency* **81** (2018), 160–171.
4. [Decision making with limited feedback: Error bounds for recidivism prediction and predictive policing](#) (with D. Ensign, S. A. Friedler, C. Scheidegger, and S. Venkatasubramanian), *Proc. Mach. Learn. Res., Algorithmic Learning Theory* **83** (2018), 359–367.
5. [Kostka numbers and longest increasing subsequences](#) (with A. Krishnan), [arXiv:1907.03881](#).
6. [Long mutation cycles](#) (with S. Fomin), *Sel. Math.* **31** (2025).
7. [On the phase diagram of the polymer model](#) (with A. Krishnan and S. Mkrtchyan), *J. Theoret. Probab.* **39** 48 (2026).
8. [Cyclically ordered quivers](#) (with S. Fomin), [arXiv:2406.03604](#), *Advances in Mathematics*.

¹Updated September 26, 2026

9. [Mutation-acyclic quivers are totally proper](#), [arXiv:2409.17832](#).
10. [Mutation cycles from reddening sequences](#) (with T. Ervin), [arXiv:2504.06573](#), Annals of Combinatorics.
11. [Mysterious points in keys but not trees](#) (with J. Simental), [arXiv:2603.17890](#).
12. [Eventual sign coherence](#) (with A. Burcroff), [arXiv:2605.12865](#).

RESEARCH AND TEACHING EXPERIENCES

University of Utah

- Research Experience for Undergraduates, November 2013–April 2016. Supervisor: Duncan Metcalfe. Created software to analyze the radiocarbon dating curve.
- Research Experience for Undergraduates, January–May 2016. Supervisor: Tommaso de Fernex. Exploring tropical calculus and the log canonical threshold.
- Research assistant, January–July 2018. Supervisor: Aditya Bhaskara. Studying robustness in high rank tensor decomposition.

University of Michigan

- Grader, Math 596: Complex Analysis (Fall 2020).
- Grader, Math 481: Introduction to Mathematical Logic (Fall 2020).
- Lab Instructor, Math 215: Multivariable and Vector Calculus (Fall 2021).
- Teaching Assistant, [Michigan Math and Science Scholars](#): Graph Theory (Summer 2022).
- Instructor, Math 105: Data, Functions, and Graphs (Fall 2019, Fall 2022).
- Instructor, Math 115: Calculus I (Winter 2020, Fall 2023).

Mathematical Reviews

- Data science consultant (Rackham internship), January–April 2024. Supervisor: Erol Ozil. Exploring machine learning approaches to automated MSC identifier assignment.

ICERM

- Semester postdoc, September–December 2025. Mentor: José Simental. For the Categorification and Computation in Algebraic Combinatorics semester program.

LACIM

- Semester postdoc, January–August 2026. Mentor: Hugh Thomas. Generalizing mutation invariants to skew-symmetrizable matrices.

CONFERENCES AND SEMINAR TALKS

- *International Conference on Algorithmic Learning Theory*, Kyoto, Japan, October 2017.
- *International Conference on Algorithmic Learning Theory*, Lanzarote, Spain, October 2018.
- *AMS Sectional Meeting: Cluster algebras, positivity and related topics*, Cincinnati, April 2023.
- *Algebraic Geometry Seminar*, UC Davis, February 2024.
- *LACIM Seminar*, UQAM, Canada, September 2024.
- *Richard P. Stanley Seminar in Combinatorics*, Harvard/MIT, September 2024.
- *Combinatorics seminar*, University of Michigan, September 2024.
- *Combinatorics seminar*, University of Minnesota, October 2024.
- *Combinatorics seminar*, University of Washington, October 2024.
- *Paris algebra seminar*, Paris, October 2024.
- *Algebra-topology seminar*, University of Alabama, November 2024.
- *UCLA Combinatorics Forum*, UC Los Angeles, December 2024.
- *AMS Sectional Meeting: Mutation and cluster algebras*, Clemson, March 2025.
- *Graduate Student Combinatorics Conference*, University of Southern California, March 2025.
- *Combinatorics and Graph Theory seminar*, Michigan State University, April 2025.
- *2025 CMS Summer Meeting*, Université Laval, June 2025.
- *ICERM Machine Learning Seminar*, Brown University, October 2025.
- *LACIM Seminar*, UQAM, January 2026.
- *Auslander Centennial Conference*, Woods Hole Oceanographic Institute, April 2026.
- *Algebraic and Enumerative combinatorics seminar*, University of Waterloo, June 2026.
- *Sumner B Myers Colloquium*, University of Michigan, September 2026.

STUDENT SEMINAR TALKS

Graduate Student Combinatorics Seminar, University of Michigan:

- February 2020. Title: *Longest increasing subsequences and Kostka numbers*.
- December 2020. Title: *The RSK algorithm for longest paths*.
- March 2021. Title: *Quiver (mutation) invariants*.
- September 2021. Title: *CC systems and vortex-free tournaments*.
- January 2022. Title: *Some colourful convexity theorems*.
- November 2022. Title: *Sets that are (NP) hard to count*.

- January 2023. Title: *RSK*.
- September 2023. Title: *Mutation graphs*.
- January 2024. Title: *Long mutation cycles*.
- September 2024. Title: *Acyclic quivers and admissible quasi-Cartan companions*.
- February 2025. Title: *The Markov equation*.

Algebraic Combinatorics Learning Seminar, University of Michigan:

- March 2022. Title: *Computing cluster cohomology*.
- February 2023. Title: *Deciding the vanishing of Schubert Polynomial Coefficients*.
- March 2024. Title: *Examples with combinatorial Chow rings*.

Pre-seminars: MIT (September 2024), Minnesota (October 2024), Washington (October 2024), ICERM (October 2025), Lacim (January 2026), Waterloo (June 2026).

Posters: Undergraduate Research Symposium at University of Utah (2016), Category Theory, Combinatorics, and Machine Learning workshop at ICERM (September 2025), Combinatorial Algebra meets Algebraic Combinatorics (January 2026).

PROFESSIONAL SERVICE AND OUTREACH

- Mentor, Directed Reading Program (2022).
- Member, [Michigan Research Experience for Graduates Committee](#) (2022).
- Event assistant, [Midwest Research Experience for Graduates](#) (2023).
- Co-organizer, [UM Student Combinatorics Seminar](#) (2021-2024).
- Member, Graduate Student Advisory Committee (2021–2025).
- Event assistant, UM branch, Association for Women in Mathematics (2023–2025).
- Co-organizer, Machine learning seminar at ICERM, (2025).
- Referee, *Electronic Journal of Combinatorics*, *Annals of Combinatorics*, *Journal of Pure and Applied Algebra*.
- Co-organizer, LEAN at LACIM (Spring 2026).
- Mentor, LACIM internships (Summer 2026).

AWARDS AND DISTINCTIONS

- Dean's List, College of Science, University of Utah (2014–2017).
- First place, Undergraduate Problem Solving Contest, Math Department, University of Utah (2015).
- Honorable mention, Mathematical Contest in Modeling (2015–2016).
- Honorable mention, Outstanding Undergraduate Researcher Award for Ph.D.-granting institutions, Computing Research Association (2018).

- Winston Churchill Scholarship (2018).
- Honored Instructor, Michigan Housing (Fall 2022).
- NSF RTG Research Assistant (Winter 2022, Winter 2023).
- Pat Shure Excellence in Teaching Award (2024).
- Rackham Predoctoral Fellowship (2024-2025).
- Sumner B. Myers award (Best Thesis, 2025).